

Performance Monitoring-Enabled Reliable AI-Based CSI Feedback

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Abstract—Artificial intelligence (AI) has emerged as a promising tool in channel state information (CSI) feedback tasks. Although current research primarily focuses on improving feedback accuracy through innovative AI approaches, the reliability of these systems in real-world scenarios often goes overlooked. Specifically, a closer examination of the feedback accuracy of individual CSI samples reveals significant variations, underscoring the imperative need for performance monitoring of AI-based CSI feedback. Building upon this observation, we introduce a pragmatic framework for AI-based CSI feedback. This process involves assessing feedback accuracy (i.e., conducting performance monitoring) on the user side before transmitting the CSI codeword. In particular, this method utilizes a lightweight proxy decoder, trained via knowledge distillation, to emulate the mapping function of the original decoder at the base station. The goal is to generate, at the user end, CSI identical to that produced at the base station by the original, more

powerful decoder, thus enable precise prediction of feedback accuracy. Simulation results demonstrate that our proposed performance monitoring method can precisely predict feedback accuracy with low complexity and accurately detect low-quality feedback samples with a detection rate of nearly 95%, ensuring reliable transmission.

Index Terms—CSI feedback, deep learning, performance monitoring, knowledge distillation.

I. INTRODUCTION

IN JUNE 2023, the International Telecommunication Union (ITU) approved the IMT-2030 Framework, a vision for 6G that outlines the framework and objectives for the future development of International Mobile Telecommunications (IMT) beyond 2030 [2]. This framework recognizes artificial intelligence (AI) as a new usage scenario, distinguishing 6G from 5G. AI's ability to process large datasets makes it invaluable for understanding propagation environments, such as channel state information (CSI) [3]. In addition to AI, immersive communication, extending enhanced mobile broadband (eMBB) from 5G, is also one of the usage scenarios in 6G. Massive multiple-input and multiple-output (MIMO) systems [4], where base stations (BSs) employ numerous transmitting antennas, are a key technology to satisfy this usage scenario [5], [6]. Despite its promise, this setup poses challenges in frequency division duplexing systems, where acquiring CSI is difficult due to the lack of channel reciprocity, as downlink and uplink transmissions operate on different frequency bands. The downlink CSI, estimated via pilot signals from the BS, must be fed back to the BS through the uplink control channel [7]. The high number of BS antennas increases feedback overhead, consuming substantial uplink resources. The CSI should be sent back with tolerable overhead and approved accuracy in accordance with system requirements. Therefore, developing efficient CSI feedback compression and reconstruction algorithms is crucial in 6G [6].

A. Related Work

Research in this area primarily focuses on two prospective approaches. First, more innovative CSI compression and reconstruction algorithms should be developed. State-of-the-art research has seen AI outperform traditional algorithms in boosting CSI feedback accuracy within certain overheads. Second, incorporating side information, e.g., correlations and multi-modal data, into the CSI feedback and reconstruction process could significantly reduce feedback overhead without sacrificing accuracy.

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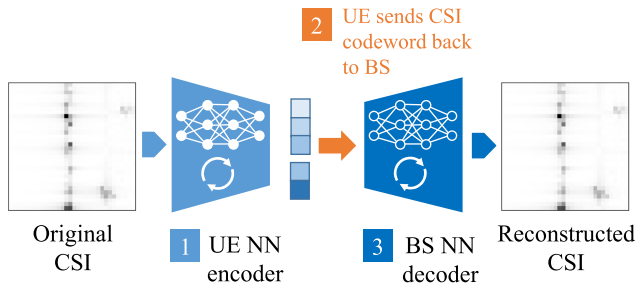


Fig. 1. AI-based autoencoder for CSI feedback, where the encoder and decoder to actualize the CSI compression and subsequent reconstruction.

The catalyst for using AI in CSI feedback enhancement was initiated by CsiNet [8], which pioneered an AI-based autoencoder framework (as shown in Figure 1) for this task. In CsiNet, a neural network (NN)-based encoder at the user equipment (UE) collaborates with a decoder at the base station (BS) to compress and reconstruct the CSI matrix. Following this framework, various studies have been conducted. For instance, some studies have improved the efficiency of AI-based feedback¹ by integrating novel NN architectures, such as attention mechanisms [9], [10], super-resolution NNs [11], and generative adversarial networks [12], and by analyzing and utilizing correlations of CSI data in the time domains [13], [14], frequency domains [15], [16], and spatial domains [17]. Moreover, CSI feedback is jointly designed with other communication modules, such as channel estimation [18] and beamforming [19], to maximize the overall performance. AI-based CSI feedback has also been investigated in emerging scenarios, such as reconfigurable intelligent surfaces [20], extremely large-scale MIMO [21], and deep data hiding [22].

Differing from the explicit CSI feedback mechanism used in CsiNet, research [23] proposed the feedback of a precoding vector, calculated from the complete downlink CSI matrix based on 5G New Radio (NR) feedback strategies, through an autoencoder. A study [24] has verified that this method significantly enhanced feedback accuracy and reduced overhead compared to the current 5G NR feedback codebooks. Additionally, AI-based feedback necessitates extensive training with CSI samples. To address this challenge, innovative solutions such as utilizing uplink CSI data for downlink feedback NN training [25] and data augmentation [26] have been proposed. Furthermore, methodologies like meta-learning and transfer learning [27] were also under exploration to expedite NN training.

From the industry side, the 3rd Generation Partnership Project (3GPP) has been exploring AI-native air interfaces in 5G-Advanced. A study item, “Study on AI/ML for New Radio (NR) air interface,” part of 3GPP Release 18 [28], focuses on three use cases: AI for CSI feedback enhancement, beam management, and position accuracy improvement. This study evaluates the potential of AI in next-generation air interfaces and aims to develop a support framework for AI-based communication algorithms. Following this, in December

2023, 3GPP approved a work item [29], “New WID on AI/ML for NR air interface.” One of the objectives of this item is to address the deployment and standardization challenges of AI-based CSI feedback enhancement. To date, numerous studies from both academic research [30] and industry reports [31] have highlighted AI’s potential to advance CSI feedback.

B. Remaining Problems

Despite these advancements, challenges persist in effectively incorporating AI into air interfaces. In particular, many studies overlook the need for additional operations after deployment. Conventional codebook-based feedback frameworks used in 5G NR are underpinned by strict mathematical proofs, ensuring consistent feedback performance across various scenarios. In contrast, AI-based CSI feedback, often perceived as a black box, depends on automated learning from training datasets and lacks this consistency. AI-based CSI feedback may face three challenges in this context:

First, generalization is a common issue. NNs trained on specific CSI sample distributions may underperform when exposed to changing propagation environments.

Second, current research mainly aims to reduce *average* errors, employing metrics such as normalized mean square error (NMSE), generalized cosine similarity (GCS), and square GCS (SGCS) to assess NN feedback accuracy [30]. However, this focus on average performance often overlooks the accuracy of individual CSI samples, potentially leading to disruptions in downlink transmission due to exhibiting low feedback accuracy in some CSI samples (see [30, Figure 34]).

Third, integrating AI-based CSI feedback into existing systems is another critical yet often overlooked aspect. Extensive research on AI-based implicit feedback [23], [26], [32], [33], aligning with current 5G NR systems as discussed in 3GPP Release 18 [31], typically neglects CSI’s essential role in system throughput estimation. They focus on compressing and reconstructing the downlink CSI’s eigenvector (precoding vector) using the UE’s encoder and the BS’s decoder. However, to design effective downlink transmission, it is imperative to relay additional channel data, such as the Channel Quality Indicator (CQI) estimated at the UE, back to the BS. The CQI assists the BS in choosing the appropriate modulation scheme and code rate, optimizing system throughput while maintaining the required block error rate [34]. The CQI is also linked to the precoding vector. Conventional codebook-based feedback systems ensure that the reconstructed vector at the BS matches that at the UE due to a one-to-one codeword-vector mapping. However, in AI-based CSI feedback, the UE is left with only the perfect precoding vector and its compressed codeword, lacking insights into the reconstructed vector at the BS. The CQI generation in AI-based CSI feedback is overlooked by current works.

Regarding the first challenge, employing a library of NN models, each suited to a particular propagation environment, is a possible solution. For the second challenge, increasing feedback bits can improve accuracy but at the cost of higher feedback overhead. For the third challenge, obtaining the reconstructed precoding vector at the UE is a way to solve the

¹In this work, the “AI-based CSI feedback” refers to the autoencoder-based CSI feedback mechanism.

problem. Overall, pre-estimating reconstructed CSI accuracy at the UE side before transmitting the compressed CSI to the BS, so that the UE can adjust its NN model adoptions or estimate a proper CQI, is necessary to resolve these challenges. Therefore, obtaining the reconstructed precoding vector at the UE is necessary for estimating downlink transmission quality. However, this approach significantly increases the complexity at the UE side due to the much higher complexity of the decoder compared to the encoder.

C. Main Focus of This Paper

To tackle these challenges, we propose a performance monitoring-enabled reliable framework for AI-based CSI feedback. This framework emphasizes accuracy monitoring before feedback due to the fluctuating performance of NNs, generalization issues, and CQI selection. The primary objective of the performance monitoring module is to enable the UE to forecast the CSI reconstruction accuracy at the BS before feedback. Only when the reconstruction accuracy meets the system requirement can the compressed CSI be fed back to the BS; otherwise, alternative strategies (sending additional feedback or switching NN/non-NN models) must be implemented. Specifically, we utilize a proxy NN model, a streamlined decoder, to reconstruct the CSI from the codeword. By aligning the proxy's output with the original decoder's and applying knowledge distillation, we can accurately predict feedback performance. Our proposed AI-based CSI feedback framework aims to ensure feedback accuracy, thus preventing poor-quality downlink transmission.

The major contributions of our work are:

- Highlighting the often-overlooked challenge of individual sample accuracy in AI-based feedback.
- Conducting detailed simulations to compare AI-based methods with traditional codebook approaches, underscoring the necessity for NN performance monitoring.
- Introducing a practical performance monitoring-enabled reliable AI-based CSI feedback framework that prioritizes the evaluation of feedback accuracy before transmission.
- Proposing a simple yet effective performance monitoring method employing a lightweight decoder, which accurately predicts feedback performance with minimal errors.

This paper is organized as follows: Section II describes the system model, including the signal model and the fundamentals of prevalent AI-based CSI feedback. Section III explores the accuracy distribution in AI-based CSI feedback. Section IV introduces our proposed performance monitoring-enabled framework for AI-based CSI feedback, along with a simple yet effective performance monitoring method. Section V presents results and discussions from numerical simulations. The paper concludes in Section VI.

D. Notation

Bold uppercase and lowercase letters represent matrices and vectors, respectively. The symbols $(\cdot)^H$ and $\|\cdot\|_2$ denote the Hermitian transpose and the L2 norm, respectively. $\mathbf{I}_N$ signifies the N -dimensional identity matrix. The major abbreviations used in this paper are summarized in Table I.

TABLE I
LIST OF MAJOR ABBREVIATIONS

Abbreviations	Definitions
CDF	Cumulative Distribution Function
CNN	Convolutional Neural Network
CQI	Channel Quality Indicator
FC	Fully Connected
GCS	Generalized Cosine Similarity
NLOS	Non-Line-Of-Sight
NMSE	Normalized Mean Square Error
NN	Neural Network
RB	Resource Block
SGCS	Square Generalized Cosine Similarity
SVD	Singular Value Decomposition
UMa	Urban Macro
UMi	Urban Micro

II. SYSTEM MODEL

A. Signal Transmission Model

This work considers a massive MIMO system that is operated on orthogonal frequency division multiplexing (OFDM) mode with N_{sb} subbands. The BS is equipped with $N_t \gg 1$ transmitting antennas, and the UE is equipped with $N_r > 1$ receiving antennas. Current 5G NR systems use a subband-based feedback approach to reduce feedback overhead. This approach involves feeding back a CSI matrix for each subband, instead of transmitting the CSI matrix for each individual subcarrier or resource block (RB). For simplicity, we focus only on the transmission and feedback within each subband moving forward.²

Assuming single-stream transmission, the signal received at the UE over the n -th subband can be written as

$$\mathbf{y}_n = \mathbf{H}_n \mathbf{v}_n x_n + \mathbf{z}_n, \quad (1)$$

where $\mathbf{H}_n \in \mathbb{C}^{N_r \times N_t}$ and $\mathbf{v}_n \in \mathbb{C}^{N_t \times 1}$ represent the downlink channel and the precoding vector over the n -th subband, respectively, and x_n and $\mathbf{z}_n \in \mathbb{C}^{N_r \times 1}$ denote the transmitted symbol and the complex additive white Gaussian noise, respectively. Singular value decomposition (SVD)-based precoding is adopted by current systems. The SVD process for the channel over the n -th subband is as follows:

$$\mathbf{H}_n = \mathbf{U}_n \mathbf{\Sigma}_n \mathbf{V}_n^H, \quad (2)$$

where $\mathbf{U}_n \in \mathbb{C}^{N_r \times N_r}$ and $\mathbf{V}_n \in \mathbb{C}^{N_t \times N_t}$ denote the unitary matrices, and the diagonal matrix $\mathbf{\Sigma}_n$ consists of non-negative real values on its diagonal and zero values elsewhere.

Given that single-stream transmission is considered, the SVD precoding vector to be fed back to the BS is the eigenvector associated with the maximum singular value in the unitary matrix $\mathbf{V}_n$ denoted as $\mathbf{v}_n$. The eigenvectors (precoding

²This model follows the one introduced in [32]; for additional details, see Section II-B of [23].

vectors) extracted from all subbands are compiled as

$$\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_{N_{sb}}] \in \mathbb{C}^{N_t \times N_{sb}}. \quad (3)$$

Reference [23] refers to the feedback of the matrix $\mathbf{V}$ as implicit CSI feedback, which is distinct from explicit CSI feedback that involves feeding back the whole CSI matrix $[\mathbf{H}_1, \dots, \mathbf{H}_{N_{sb}}]$ to the BS. If $\mathbf{V}$ is fed back to the BS without compression or quantization and each real value is quantized with 16 bits, the feedback bit number is $32N_tN_{sb}$. For example, assuming N_t and N_{sb} to be 32 and 13, respectively, the feedback bit number is 13,312. This substantial feedback overhead requires extensive uplink transmission resources and is impractical in real-world systems. Therefore, compressing and quantizing the matrix $\mathbf{V}$ is crucial.

B. AI-Based CSI Feedback

In AI-based CSI feedback, an autoencoder, composed of an encoder and a decoder, is utilized to compress and reconstruct the matrix $\mathbf{V}$. Considering that both the UE and the BS are equipped with an NN module, the 3GPP report [35] describes the autoencoder for CSI feedback as a two-sided model.

Specifically, upon obtaining the downlink CSI from the received pilot signals, the UE extracts the precoding matrix $\mathbf{V}$ from $[\mathbf{H}_1, \dots, \mathbf{H}_{N_{sb}}]$. It then compresses and quantizes the extracted precoding matrix using the NN-based encoder as follows:

$$\mathbf{s} = \mathcal{Q}(\mathbf{f}_{\text{en}}(\mathbf{V}; \boldsymbol{\Theta}_{\text{en}})), \quad (4)$$

where $\mathbf{f}_{\text{en}}(\cdot)$ and $\mathcal{Q}(\cdot)$ denote the compression operation (encoder) and the quantization operation, respectively, with $\boldsymbol{\Theta}_{\text{en}}$ being the corresponding NN parameters of the encoder. The encoder typically consists of two main components: feature extraction and dimensionality reduction. Feature extraction is usually conducted using convolutional NNs (CNNs), while dimensionality reduction often involves reducing the number of neurons in the fully connected (FC) layer. Uniform quantization is the commonly employed strategy in AI-based CSI feedback [30].

Subsequently, the BS reconstructs the precoding matrix $\mathbf{V}$ based on the quantized codeword as:

$$\hat{\mathbf{V}} = \mathbf{f}_{\text{de}}(\mathbf{s}; \boldsymbol{\Theta}_{\text{de}}), \quad (5)$$

where $\mathbf{f}_{\text{de}}(\cdot)$ and $\boldsymbol{\Theta}_{\text{de}}$ represent the reconstruction operation (decoder) and the corresponding NN parameters, respectively. $\hat{\mathbf{V}}$ symbolizes the reconstructed stacked precoding matrix, i.e., $[\hat{\mathbf{v}}_1, \dots, \hat{\mathbf{v}}_{N_{sb}}]$. The NN-based decoder consists of two modules: dimensionality increase and refinement. The first module typically increases the number of neurons in the FC layer, followed by several CNN layers to refine the initially reconstructed CSI.

The encoder and decoder are jointly trained to maximize the GCS ρ or the SGCS ρ^2 . The SGCS ρ^2 between $\mathbf{V}$ and $\hat{\mathbf{V}}$ is defined as:

$$\rho^2 = \frac{1}{N_{sb}} \sum_{n=1}^{N_{sb}} \left(\frac{|\mathbf{v}_n^H \hat{\mathbf{v}}_n|}{\|\mathbf{v}_n\|_2 \|\hat{\mathbf{v}}_n\|_2} \right)^2. \quad (6)$$

Moving forward, for clarity, let us denote ρ_i^2 as the SGCS value for the i -th CSI sample and ρ^2 as the average SGCS across all samples. In this study, SGCS is specifically adopted as the optimization objective. Accordingly, the loss function is defined as:

$$L_{\text{SGCS}} = 1 - \rho^2. \quad (7)$$

The end-to-end training is executed using gradient descent-based backpropagation, which involves computing gradients and iteratively updating the model parameters. Notably, the quantization operation is non-differentiable, posing a challenge for backpropagation. This issue is addressed by setting the gradient of the quantization operation to one [23], [32].

The creation of CQI necessitates an initial estimation of the signal-to-noise ratio (SNR) using implicit CSI as a precoder, followed by effective SNR (eSNR) mapping and table lookup [36]. As a result, CQI is selected and fed back to the BS along with the Precoding Matrix Indicator (PMI) by the UE, enabling the BS to adjust data streams with the appropriate modulation scheme, code rate, and precoder. The SNR is essential in the CQI generation, and the SNR (ρ_n) for the n -th subband is calculated as:

$$\rho_n = \frac{\|\mathbf{H}_n \hat{\mathbf{v}}_n\|_2^2}{\sigma^2}, \quad (8)$$

where σ^2 denotes the variance of the additive white Gaussian noise. In codebook-based CSI feedback, the UE and BS share a codebook. Searching for the codeword can be time-consuming, but retrieving it using a feedback index is fast and concise, allowing the UE to quickly obtain the reconstructed vector $\hat{\mathbf{v}}_n$ from the codebook. Upon acquiring the vector $\hat{\mathbf{v}}_n$, the UE proceeds to calculate the SNR according to (8) and then generate CQI.

III. KEY OBSERVATIONS

In autoencoder-based implicit CSI feedback, the primary training objective is minimizing the average feedback errors, as indicated in (7). Consequently, the average SGCS is commonly utilized to evaluate the accuracy of AI-based CSI feedback. However, this evaluation method may overlook the accuracy of individual samples. Given the inherent instabilities in AI systems [37], [38], NNs might exhibit subpar performance on certain samples. Unfortunately, due to the averaging process, these instances of poor performance are often overlooked, potentially leading to low-quality downlink transmission in practical systems. On the other hand, the feedback NN may underperform when the training and test CSI data distributions are mismatched, resulting in a diminished average feedback accuracy. This implies poor generalizability of AI-based CSI feedback. Nevertheless, to date, no studies have assessed the feedback accuracy of individual CSI samples when there is a distribution mismatch within NNs. In this section, we present key observations from simulation experiments to substantiate the above arguments. These observations underpin the motivation behind the proposed performance monitoring-enabled AI-based CSI feedback framework.

TABLE II
FEEDBACK PERFORMANCE COMPARISON BETWEEN THE AI-BASED CSI FEEDBACK AND TYPE II CODEBOOK

Metric Method	Average performance		Individual performance		
	Feedback bit number (N_{bits})	SGCS (ρ^2)	Standard deviation (σ)	Variance (σ^2)	Coefficient of variation (CV)
Type II codebook	160	0.8032	0.1068	0.0114	0.1330
AI-based CSI feedback	112	0.8015	0.1393	0.0194	0.1737

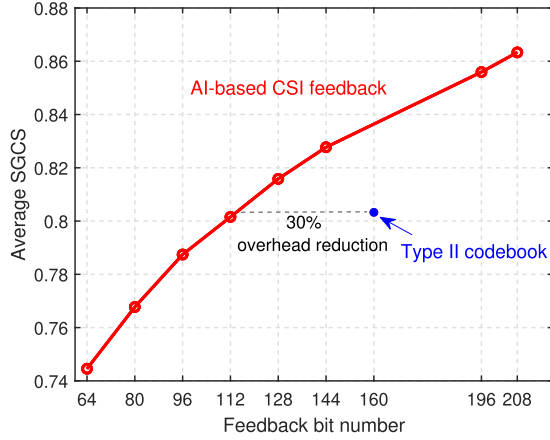


Fig. 2. Average SGCS ρ^2 versus feedback bit number N_{bits} . AI-based CSI feedback achieves comparable performance with a 30% overhead reduction compared with the Type II codebook.

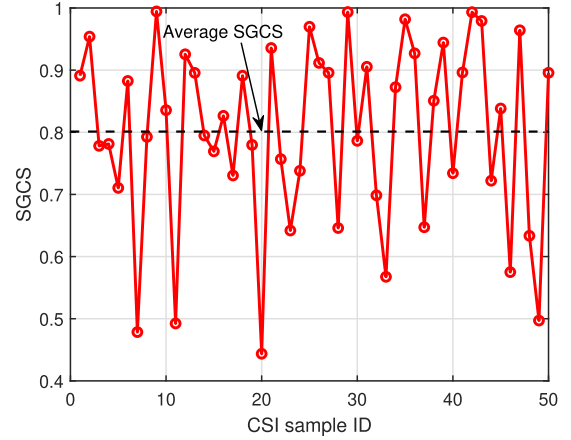


Fig. 3. SGCS for 50 distinct CSI samples using the AI-based feedback method with $N_{\text{bits}} = 112$. This figure highlights the considerable variability of AI-based CSI feedback in the performance of individual CSI samples.

In line with the consensus reached in 3GPP [35], SGCS is employed as the metric for evaluating CSI feedback performance. To this end, we train a feedback NN using a 3GPP channel dataset (Dataset A) to maximize the mean SGCS. We then evaluate the performance of the trained NN on various samples, adhering either to the same or different distributions as the training dataset. As this section focuses on the observations of AI-based CSI feedback, the architecture of the feedback NN and the parameters used for channel generation and NN training are detailed in Sections IV-B and V-A, respectively. Furthermore, we analyze both the average SGCS of the AI-based CSI feedback and the SGCS of each individual CSI sample, comparing these results with those obtained using the Type II codebook adopted in 5G NR.

A. Feedback Performance

Figure 2 illustrates the average SGCS performance for both AI-based CSI feedback and the Type II codebook across different feedback bit numbers N_{bits} . It is observed that the feedback performance improves as the number of feedback bits increases. As reported in [23] and [34], the average SGCS for the Type II codebook is 0.8032 with a feedback bit number of 160. In contrast, the average SGCS for AI-based CSI feedback is 0.8015 when the feedback bit number is reduced to 112. This demonstrates that AI-based CSI feedback achieves comparable performance with a 30% reduction in overhead, highlighting the significant potential of AI in the context of CSI feedback tasks. Additionally, if the feedback NN architecture is enhanced with more advanced NN modules, there could be an even greater improvement in performance.

B. Feedback Instability

Considering the comparable performance of AI-based CSI feedback with 112 bits to that of the Type II codebook, this configuration will be the focus for assessing SGCS in individual CSI samples. To illustrate the dispersion of SGCS for each CSI sample around the average SGCS, metrics such as standard deviation σ , variance σ^2 , and coefficient of variation CV are employed. These metrics are defined as:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (\rho_i^2 - \rho^2)^2, \quad (9)$$

$$\text{CV} = \frac{\sigma}{\rho^2}, \quad (10)$$

where N denotes the total number of testing CSI samples.

Table II presents a comparison of performance metrics (including standard deviation, variance, and coefficient of variation) between AI-based CSI feedback and the Type II codebook. Despite the similarity in average SGCS performance between AI-based and Type II codebook-based feedback methods, the standard deviation, variance, and coefficient of variation for AI-based CSI feedback are significantly higher than those for the Type II codebook. Specifically, the standard deviation of AI-based feedback is 30% higher, its variance is 70% higher, and its coefficient of variation is 30% higher than those of the Type II codebook. These findings highlight the performance instability of AI-based CSI feedback.

Furthermore, to intuitively demonstrate this performance instability, we randomly select 50 samples from the testing dataset and assess the feedback performance on each. Figure 3 displays the SGCS performance for each CSI sample. The average SGCS of these 50 samples is 0.8011, similar to that

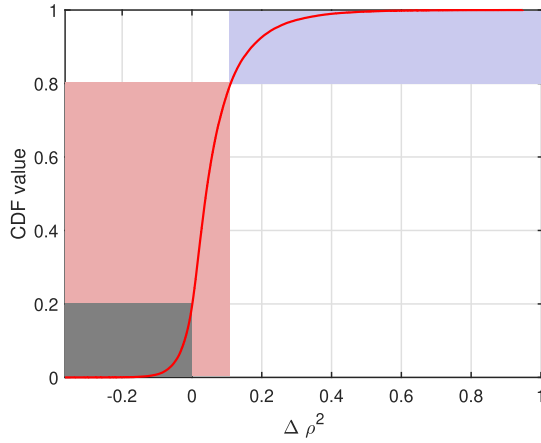


Fig. 4. The CDF of the SGCS drop $\Delta\rho_i^2$ for each CSI sample due to distribution mismatch. This illustrates that over 80% of the CSI samples experience an SGCS drop of less than 0.1.

achieved by the Type II codebook. However, it is crucial to note that there is considerable variability in the performance of individual CSI samples. Notably, the SGCS values for four CSI samples (8% of the total) are below 0.5, and eleven samples (22%) have values under 0.7. If these CSI samples were to be fed back to the BS, the downlink transmission performance could significantly deteriorate.

C. Feedback Generalization

In this subsection, we consider the case where a distribution mismatch exists between the training and test CSI datasets. Specifically, the baseline NN is still trained with the CSI samples from Dataset A but assessed using a dataset with a different distribution. The test CSI samples are randomly selected from Dataset B; more details about this dataset can be found in Section V-A.1. The simulation results indicate that, without a distribution mismatch, the average SGCS for the test CSI samples is 0.6931. However, with a distribution mismatch, this value decreases to 0.6171, demonstrating the limited generalizability of AI-based CSI feedback. Although some works have focused on improving the generalizability of AI-based CSI feedback with online learning (e.g., transfer learning [27]), no thorough analysis of the generalization performance is given. The poor generalizability is merely evident in the decreased average feedback accuracy metrics like SGCS and NMSE. This raises a pivotal question: *Do the majority of samples exhibit poor performance in the presence of a distribution mismatch?*

Figure 4 plots the cumulative distribution function (CDF) value of the feedback SGCS drop $\Delta\rho_i^2$ for each CSI sample arising from distribution mismatch. A positive $\Delta\rho_i^2$ indicates that the NN trained with matched CSI samples outperforms the one trained with mismatched CSI samples, and vice versa. This figure reveals that over 80% of CSI samples exhibit an SGCS drop below 0.1, with nearly 20% even showing an increase in feedback accuracy under the mismatched scenario. Focusing solely on these samples, the average SGCS drop is a mere 0.0262, which appears to be practically inconsequential. However, considering the remaining 20% of samples, these samples demonstrate

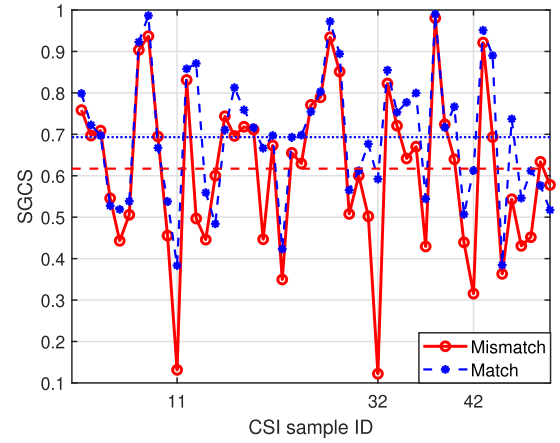


Fig. 5. SGCS for 50 different CSI samples using the AI-based feedback method with $N_{\text{bits}} = 112$. This figure highlights the impact of a distribution mismatch between the training and test datasets on the feedback performance.

significant degradation in feedback accuracy, leading to concerns about the generalization capabilities of AI-based CSI feedback.

Moreover, similarly to Figure 3, we randomly select 50 samples and assess the feedback performance of the NNs trained with matched and mismatched datasets on each selected CSI sample. From Figure 5, first, both the matched and mismatched NNs demonstrate significant instability. Second, the SGCS gaps $\Delta\rho_i^2$ of most samples are small. When the mismatched NN is adopted, the three CSI samples (sample IDs: 11, 32, and 42) have the lowest SGCS values, which are 0.1315, 0.1221, and 0.3154, respectively. The corresponding SGCS gaps of these three CSI samples are 0.2518, 0.4700, and 0.2972, respectively. In practical communications, the system cannot operate effectively with such low-quality CSI.

Consequently, in efforts to enhance generalization, it is crucial to focus more attention on these specific samples rather than treating all samples equally. Thus, identifying these samples through performance monitoring is essential. Moreover, if a matched NN is available, the UE just needs to switch to the matched NN; otherwise, the UE has to feed back these samples with non-AI methods (e.g., Type II codebook) and then store these samples for future online training.

IV. PROPOSED PERFORMANCE MONITORING-ENABLED RELIABLE AI-BASED CSI FEEDBACK FRAMEWORK

From the discussions in the previous section, it is clear that implementing an efficient performance monitoring mechanism is critical. This mechanism should be designed to predict feedback accuracy in advance, thus ensuring transmission reliability. However, the integration of AI with communication systems for performance monitoring is not widely explored in conventional communication algorithms. Therefore, a specialized framework must be developed with great care to support AI-based CSI feedback in practical deployments [35]. Prior to introducing the baseline feedback autoencoder and proposing an effective performance monitoring method, we will first explore a performance monitoring-enabled practical framework for AI-based CSI feedback in the subsequent subsection.

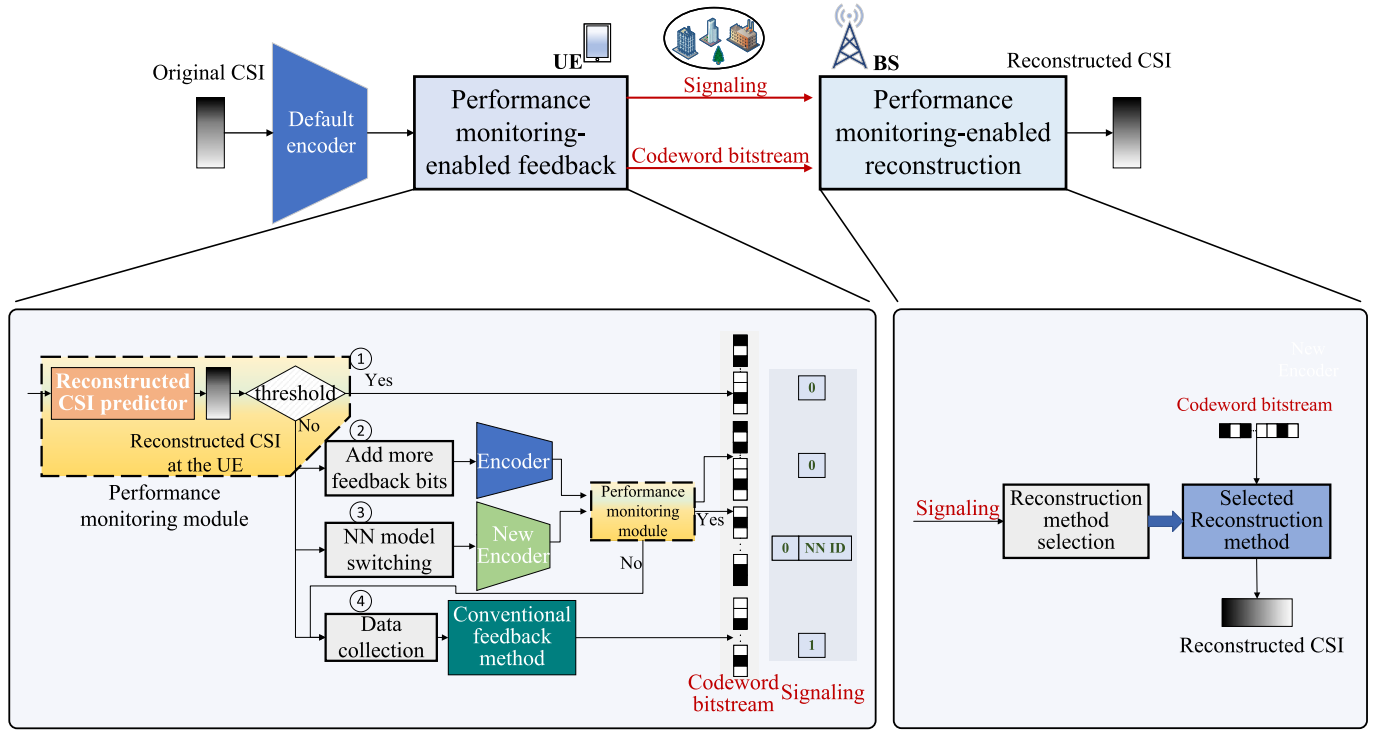


Fig. 6. The proposed performance monitoring-enabled framework for reliable AI-based CSI feedback in practical scenarios, aimed at preventing substantial performance decline due to low-quality CSI feedback.

A. Practical Performance Monitoring-Enabled Framework for AI-Based CSI Feedback

Let us now examine a practical framework for AI-based CSI feedback. As depicted in Figure 6, this proposed framework is designed to mitigate significant performance degradation due to low-quality CSI feedback and to ensure reliable downlink transmission. The framework includes the following key steps. First, the UE compresses and quantizes the downlink CSI $\mathbf{V}$ using the default encoder. Then, the output of the default encoder is sent to the performance monitoring-enabled feedback module, which feeds back the codeword and signaling bitstreams to the BS according to the feedback performance predicted at the UE. Specifically, if the feedback accuracy meets the system requirement (e.g., SGCS threshold t_{SGCS}), the original codeword is transmitted to the BS. If the feedback accuracy falls below t_{SGCS} , alternative strategies³ are employed, and signaling, indicating which type of compression (encoder) method is adopted, is also generated. A signaling design example is given in Figure 6. Once the BS receives the feedback bitstreams, including the codeword and signaling, a performance monitoring-enabled reconstruction process is employed, in which the corresponding reconstruction (decoder) method is selected according to the signaling bitstream, and then the downlink CSI is recovered based on the codeword.

The alternative strategies mentioned before might involve the following two scenarios:

- **More feedback bits are needed:** As shown in Figure 3, due to the instabilities of AI-based CSI feedback, the accuracy of some CSI samples may be significantly lower than others. Thus, allocating more feedback bits⁴ to enhance their accuracy becomes necessary.
- **The feedback NN model is mismatched:** As discussed in Section III-C, feedback accuracy will considerably decrease if there is a mismatch between the training and practical CSI distributions.
 - If the UE has stored an NN model that matches the real-world environment, it switches to the appropriate feedback model, then transmits the NN model's ID to the BS. The BS reconstructs the downlink CSI using the designated decoder based on the received feedback codeword.
 - If the environment is entirely new, the UE stores the CSI samples and transmits them to the BS for online training during idle periods. In this case, conventional feedback algorithms, e.g., Type II codebook, are used for feedback of the downlink CSI.

To avoid/mitigate the potential negative effects of unsuitable changes in the feedback strategy, the communication scenario should be taken into account. In communication scenarios demanding high reliability, such as autonomous driving, directly switching to the conventional feedback strategy (e.g., TYPE II codebook) significantly mitigates associated risks but does so at the expense of more feedback bits. In other communication scenarios, if the newly adopted feedback

³Given the objective of this study to introduce a reliable AI-based CSI feedback framework supported by performance monitoring, only a basic approach related to feedback strategy change is outlined at this point.

⁴Storing different NNs for varying bit feedbacks needs large storage at the UE. The multi-rate encoder/decoder framework in [10] and [39] can help solve this.

strategy fails to meet the requirement, the conventional feedback strategy is directly implemented. Moreover, the SGCS threshold t_{SGCS} is a key hyperparameter. Selecting a suitable SGCS threshold t_{SGCS} is essential. To minimize the overhead of CSI feedback while maintaining satisfactory SGCS performance (that is, the SGCS threshold t_{SGCS}), it is essential to consider the specific communication scenarios. For instance, in single-user scenarios, current 5G NR systems favor low-quality CSI feedback due to its significantly reduced feedback overhead. However, in multi-user environments where user interference is a factor, the preference shifts towards high-quality feedback, even though it comes with a larger feedback overhead. Hence, in the single-user scenario, the SGCS threshold t_{SGCS} can be set comparatively low, while it is relatively higher in the multi-user scenario.

This framework supports highly reliable CSI feedback, ensuring high-quality downlink transmission. However, accurately predicting feedback accuracy at the UE, i.e., performance monitoring, remains a major challenge. The natural approach is to use the same decoder as the BS to reconstruct the downlink CSI $\mathbf{V}$ from the codeword before feedback transmission. However, as noted in Section IV-B.2, the decoder's complexity is significantly higher than the encoder's due to the greater computational resources available at the BS. If the UE directly uses the original, powerful decoder for reconstructing the CSI, two main issues arise:

- The computational resources at the UE, particularly for some low-cost devices, are limited and may not support the inference of the complex decoder.
- Even if the UE has sufficient computational resources for inference, the complex NN inference process introduces substantial latency, which is unacceptable in practical systems.

Therefore, using the same decoder for performance monitoring is not feasible in practical systems. The challenge is in crafting a new decoder that presents performance *similar* to the original one while simultaneously showing drastically reduced complexity. This can be formulated as:

$$\arg \min_{f'_{\text{de}}(\cdot; \Theta'_{\text{de}})} \|f'_{\text{de}}(\mathbf{s}; \Theta'_{\text{de}}) - \hat{\mathbf{V}}\|_2^2, \quad (11)$$

where $f'_{\text{de}}(\cdot)$ and Θ'_{de} denote the new CSI reconstruction decoder and its corresponding NN parameters, respectively. This new decoder comes with significantly less complexity than the original, f_{de} .

B. Baseline NN Architecture

The Transformer architecture [40] has become popular in both computer vision and natural language processing, demonstrating significant potential in CSI feedback [41]. In light of this advancement, we adopt a Transformer-based NN architecture in this work, as illustrated in Figure 7, serving as the *baseline* feedback NN architecture. In the following sections, we will introduce the encoder framework, the decoder framework, and the backbone Transformer block, respectively.

1) *Encoder*: In the UE, the encoder comprises three main components: feature extraction, compression, and quantization. The input shape of the implicit CSI, denoted

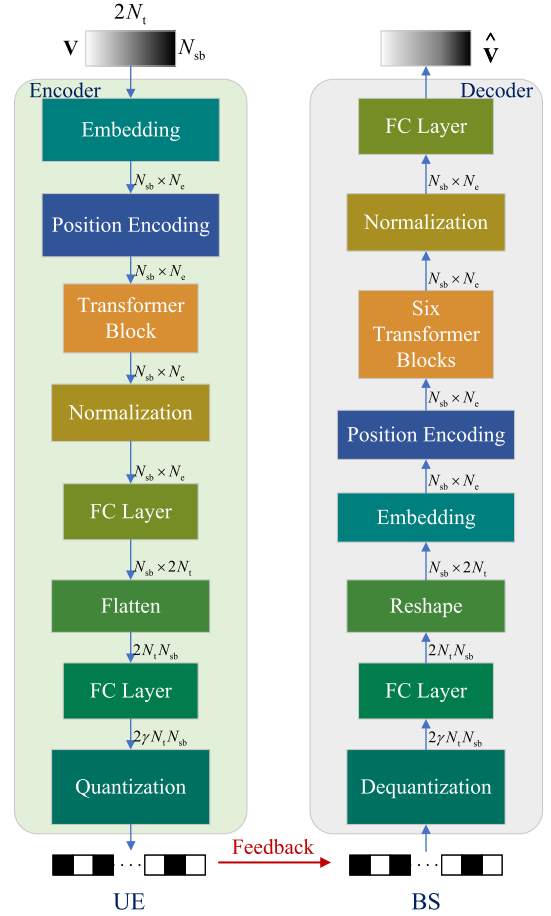


Fig. 7. Illustration of the baseline Transformer-based feedback NN architecture. The encoder and the decoder at the UE and BS compress and reconstruct the downlink CSI $\mathbf{V}$, respectively. The complexity of the decoder is much higher than that of the encoder.

as $\mathbf{V}$, is set to $N_{\text{sb}} \times 2N_t$. This format is chosen because NN libraries like TensorFlow cannot process complex values directly, and the real and imaginary parts of the complex matrix are stacked in the last dimension. Prior to entering the Transformer block, the input undergoes preparation through an embedding layer with an embedding dimension of N_e and a position encoding layer based on a learnable position vector. This is followed by a Transformer block [41], succeeded by a normalization layer and an FC layer, which operate sequentially on the last dimension to extract the implicit CSI features.

The CSI feature matrix is subsequently flattened into a vector that includes $2N_t N_{\text{sb}}$ elements. Assuming a compression ratio of $\gamma < 1$, an FC layer with $2\gamma N_t N_{\text{sb}}$ neurons compresses the downlink CSI from $2N_t N_{\text{sb}}$ to $2\gamma N_t N_{\text{sb}}$. For digital transmission purposes, the compressed CSI undergoes quantization with a uniform quantizer, denoted by the quantization bit number B . Consequently, the feedback CSI undergoes quantization with a uniform quantizer, denoted by the quantization bit number, or feedback overhead, is calculated as:

$$N_{\text{bits}} = 2\gamma N_t N_{\text{sb}} \times B. \quad (12)$$

The resulting bitstream, containing N_{bits} bits, is then transmitted back to the BS via the uplink control link.

2) *Decoder*: Upon receiving the feedback bits, the BS de-quantizes the feedback codewords. Initially, an FC layer

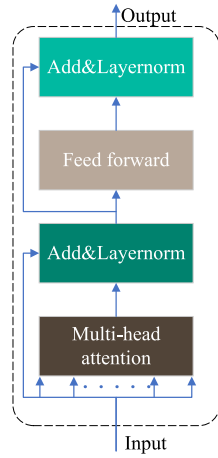


Fig. 8. Illustration of the backbone Transformer block [40], [41].

with $2N_t N_{sb}$ neurons is used to reconstruct the implicit CSI. This reconstructed implicit CSI is then reshaped to match the encoder's input size, transitioning from $1 \times 2N_t N_{sb}$ to $N_{sb} \times 2N_t$. As in the encoder, both an embedding layer and a position encoding layer are applied. Subsequently, six sequential Transformer blocks⁵ are utilized to refine the initially reconstructed implicit CSI. Finally, a normalization layer normalizes the refined implicit CSI, followed by an FC layer operating on the last dimension, which yields the final implicit CSI matrix, denoted as $\hat{\mathbf{V}}$. Owing to its much deeper NN architecture, the decoder's complexity is significantly higher than that of the encoder.

3) *Transformer Block*: The core architecture of the basic Transformer block is depicted in Figure 8. The fundamental idea behind the Transformer is self-attention. Initially, the input undergoes processing with a multi-head attention NN layer (with N_{head} heads), followed by a shortcut connection from the input and a normalization layer [40]. Subsequently, the output is passed through a feedforward NN consisting of two FC layers. A shortcut connection is also established, and the output is normalized. The normalized matrix represents the final output of the Transformer block. Due to page limitations and the focus of this paper, we refer readers to [40] and [41] for more comprehensive details regarding the basic Transformer block.

C. An Effective Performance Monitoring Method

To support the performance monitoring-enabled reliable CSI feedback framework described earlier, we propose a concise yet effective performance monitoring method. In this subsection, we first introduce the key framework, which includes the architecture and optimization strategy. Then, we address the issue of intellectual property protection in practical systems by discussing the collaborative training between UE and BS manufacturers. Finally, we introduce three evaluation metrics for performance monitoring.

1) *Key Framework*: To address the aforementioned challenges, a proxy NN model is needed, one that performs

⁵Given the powerful computational processors at the BS, we adopt more Transformer blocks to achieve superior reconstruction performance, albeit at a higher complexity [30].

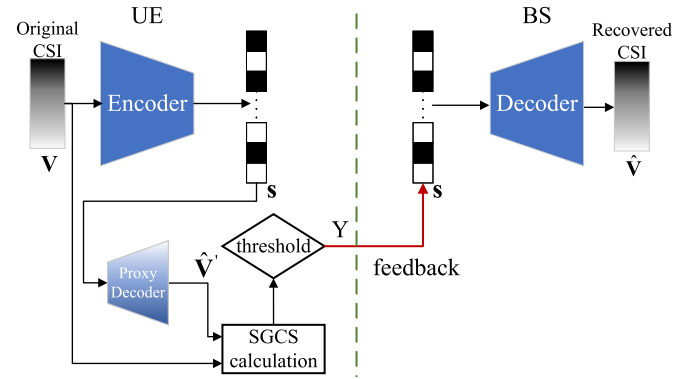


Fig. 9. The proxy-based performance monitoring framework for AI-based CSI feedback, assuming that the feedback accuracy meets the system requirements. This framework features a proxy decoder designed to mirror the functionality of the original decoder at the BS.

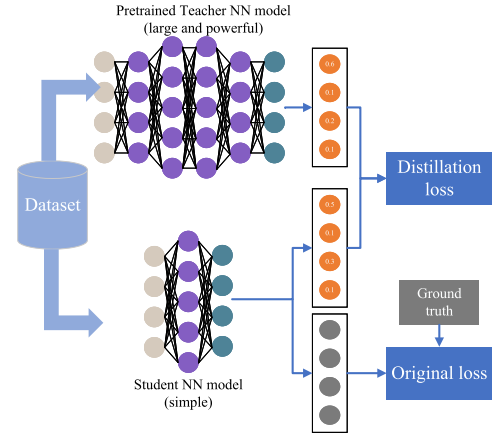


Fig. 10. An illustration of knowledge distillation, wherein the teacher model transfers knowledge to the student model.

similarly to the decoder at the BS, but with significantly lower complexity. The primary framework of our proposed proxy-based performance monitoring for AI-based CSI feedback is presented in Figure 9. After compressing the implicit channel $\mathbf{V}$ into a bitstream s , the UE recovers the initial channel using a lightweight proxy decoder. If the reconstruction accuracy meets the system requirements, the bitstream is promptly fed back to the BS. Otherwise, additional feedback bits must be supplied, or the feedback NN model should be switched, or conventional algorithms can be employed. Therefore, the primary challenge is generating a lightweight proxy decoder that performs similarly to the original decoder.

As shown in Figure 7, six Transformer blocks are adopted in the decoder to achieve high-quality reconstruction, albeit at the expense of computational complexity. In the proxy decoder, only one Transformer block is utilized, resulting in significantly lower complexity compared to the original decoder. The primary objective of the proxy model is not to reconstruct the implicit channel as accurately as possible but to mimic the mapping function of the original decoder.

Knowledge distillation [42], widely used for compressing NN models, presents a potential solution for creating the proxy decoder. The core concept involves transferring knowledge learned by the pre-trained large and powerful teacher model to the lightweight student model. Specifically, as shown in Figure 10, the training objective of the student NN model

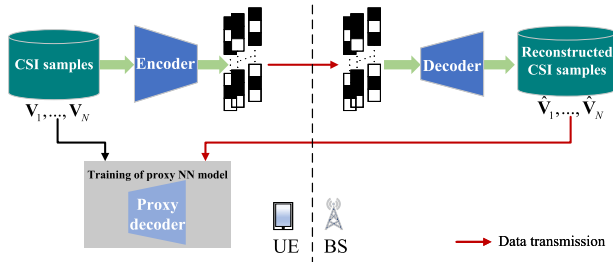


Fig. 11. Illustration of the training collaboration between the UE and BS (manufacturers), with the assumption that the proxy NN model is designed and trained by UE manufacturers.

is not solely to minimize the discrepancy between its output and the ground truth (label), but also to reduce the divergence between the outputs of the teacher and student NN models. Consequently, the final loss comprises both the original loss and the distillation loss. The knowledge learned by the teacher model is in the form of its output, and knowledge transfer between the teacher and student models is accomplished by minimizing the divergence between their outputs. In [43] and [44], to protect intellectual property and reduce NN complexity, knowledge distillation is applied to transfer the knowledge from the original encoder to a new one.

Inspired by [43] and [44], we consider the original powerful decoder at the BS as the teacher model and the lightweight proxy decoder at the UE in this work as the student model. Therefore, the main task is to transfer the knowledge learned by the original decoder to the lightweight proxy decoder. During the training process, the label for the proposed proxy decoder is set as the output of the original decoder, denoted as $\hat{V}$. Thus, the optimization problem can be formulated as follows:

$$\begin{aligned} \min_{\Theta_{de}^{proxy}} \quad & \|\hat{V} - \hat{V}'\|_2^2, \\ \text{s.t.} \quad & \hat{V}' = f_{de}^{proxy}(s; \Theta_{de}^{proxy}), \end{aligned} \quad (13)$$

where $f_{de}^{proxy}(\cdot)$ represents the proxy decoder, Θ_{de}^{proxy} represents its corresponding NN parameters, and $\hat{V}'$ represents the output of the proxy decoder. The training objective is to learn a mapping function, $f_{de}^{proxy}(\cdot)$, that maps s to the imperfect implicit channel $\hat{V}$.

The utilization of knowledge distillation to generate a proxy decoder is concise. The method's concise design, coupled with its low computational overhead, makes it a practical and efficient solution for feedback accuracy prediction in practical scenarios.

2) *Training Collaboration*: Intellectual property protection is a crucial consideration in AI-based CSI feedback [35], [43]. Unlike most communication modules, such as channel estimation and signal detection, CSI feedback involves both the UE and BS sides. Typically, the UE and BS are manufactured by different companies and cannot share detailed NN architectures due to intellectual property reasons. Consequently, training a proxy NN model without sharing NN information between the UE and BS sides presents a

significant challenge.⁶ Since the proxy model is deployed at the UE, we initially assume that the proxy NN model in this work is designed and trained by UE manufacturers.

Algorithm 1 Training of the Proxy NN Model

- 1: **Input:** Implicit CSI $\{V_1, \dots, V_N\}$
- 2: **Output:** Trained proxy NN model
- 3: **UE:**
- 4: • Collect the implicit CSI $\{V_1, \dots, V_N\}$
- 5: • Generate codewords $\{s_1, \dots, s_N\}$ using pre-trained encoder
- 6: • Transmit codewords $\{s_1, \dots, s_N\}$ to BS via uplink channel
- 7: **BS:**
- 8: • Receive codewords $\{s_1, \dots, s_N\}$ from UE
- 9: • Generate reconstructed CSI $\{\hat{V}_1, \dots, \hat{V}_N\}$ using pre-trained decoder
- 10: • Transmit reconstructed CSI $\{\hat{V}_1, \dots, \hat{V}_N\}$ to UE manufacturer
- 11: **UE (or its manufacturer):**
- 12: • Receive reconstructed CSI $\{\hat{V}_1, \dots, \hat{V}_N\}$ from BS
- 13: **If** reconstructed CSI is sent to UE manufacturer:
- 14: • Train proxy NN model with (13) using $\{s_1, \dots, s_N\}$ and $\{\hat{V}_1, \dots, \hat{V}_N\}$ on edge or central servers
- 15: • Transmit trained NN parameters to UEs
- 16: **If** reconstructed CSI is sent directly to UE:
- 17: • Train proxy NN model with (13) using novel lightweight online training strategies based on available computational power at the UE
- 18: **end**

Figure 11 illustrates the training collaboration between the UE and BS manufacturers. The training of the proxy NN model, as shown in Algorithm 1, involves the following steps:

- The UE collects the implicit CSI $\{V_1, \dots, V_N\}$ and generates the corresponding codewords $\{s_1, \dots, s_N\}$ using the pre-trained encoder.
- The UE transmits the codewords $\{s_1, \dots, s_N\}$ to the BS via the uplink channel during idle time.
- The BS generates the reconstructed CSI $\{\hat{V}_1, \dots, \hat{V}_N\}$ using the pre-trained decoder once it receives the codewords transmitted by the UE.
- The BS then transmits the reconstructed CSI $\{\hat{V}_1, \dots, \hat{V}_N\}$ to the UE (or its manufacturer).
- Finally, if the reconstructed CSI $\{\hat{V}_1, \dots, \hat{V}_N\}$ is sent to the UE manufacturer, the UE manufacturer trains a proxy NN model using $\{s_1, \dots, s_N\}$ and $\{\hat{V}_1, \dots, \hat{V}_N\}$ on edge or central servers and then transmits the NN parameters to the UEs. If the reconstructed CSI $\{\hat{V}_1, \dots, \hat{V}_N\}$ is sent directly to the UE, the UE trains the proxy NN model using novel lightweight online training strategies based on the available computational power at the UE.

⁶It is also essential to study how to separately train the encoder and decoder at the UE and BS sides. For the purposes of this work, we assume that the original encoder and decoder have already been trained and deployed at the UE and BS, respectively.

Moreover, if the proxy NN model is designed and trained by the BS manufacturer and then distributed to each UE, the training process becomes much simpler. In this case, the reconstructed CSI $\{\hat{\mathbf{V}}_1, \dots, \hat{\mathbf{V}}_N\}$ no longer needs to be transmitted to the UE (or the UE manufacturer). The BS, typically equipped with ample computational power, trains the proxy NN model using the received codewords and the CSI reconstructed by itself as the input and output labels. Once the training is completed, the BS transmits the proxy NN model architecture and its parameters to all UEs.

3) *Evaluation Metrics for Performance Monitoring*: Given that few works focus on performance monitoring of AI-based methods, evaluating the performance of such monitoring using existing metrics, such as average SGCS, proves challenging. Therefore, this section proposes three new evaluation metrics.

- The first evaluation metric is the average gap between the real and predicted accuracy. The capacity of an NN is positively correlated with its complexity, including the number of layers and other architectural intricacies [45]. While knowledge distillation enables the lightweight proxy decoder to mimic the mapping function of the original decoder, the proxy's performance is inevitably inferior due to the substantial gap in NN complexity. The average decrease in accuracy can be calculated using

$$\overline{\Delta\rho^2} = \frac{1}{N} \sum_{i=1}^N (\rho_i^2 - \hat{\rho}_i^2), \quad (14)$$

where ρ_i^2 and $\hat{\rho}_i^2$ represent the real and predicted (proxy) SGCS values of the i -th sample, respectively. During feedback accuracy prediction, the final SGCS is obtained by adding the average decrease $\overline{\Delta\rho^2}$ to the predicted (proxy) SGCS value $\hat{\rho}_i^2$. This adjustment compensates for the prediction bias resulting from the use of a significantly more streamlined decoder. Therefore, the average gap G_{ρ^2} between the real and predicted accuracy is defined as:

$$G_{\rho^2} = \frac{1}{N} \sum_{i=1}^N \left| \rho_i^2 - (\hat{\rho}_i^2 + \overline{\Delta\rho^2}) \right|. \quad (15)$$

A smaller value of G_{ρ^2} indicates a higher accuracy in the proxy model's predictions.

- The second evaluation metric is classification performance. As mentioned in Section IV-A, one of the primary purposes of performance monitoring is to determine whether the reconstructed CSI $\hat{\mathbf{V}}$ can satisfy the accuracy requirement, thus avoiding a significant drop in system performance. Consequently, this can be regarded as a binary classification task, identifying samples of CSI reconstruction that don't meet the system's requirements. If the sample SGCS value is higher than t_{SGCS} , the classification label is "True"; otherwise, it is "False." In Figure 3, if the SGCS threshold t_{SGCS} is set to 0.6, six samples (i.e., 12%) are below this threshold, resulting in "False" labels. This situation creates an imbalanced classification problem; that is, even if all prediction results are set as "True," the classification accuracy can still achieve up to 88%. Therefore, precision and

recall [46] need to be adopted to assess the classification performance.

Precision $P \in [0, 1]$ and recall $R \in [0, 1]$ are defined as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (16)$$

$$R = \frac{\text{TN}}{\text{TN} + \text{FN}}, \quad (17)$$

where TP represents the number of true positives (correct predictions when the SGCS values are higher than t_{SGCS}), FP represents the number of false positives (incorrect predictions when the SGCS values are higher than t_{SGCS}), TN represents the number of true negatives (correct predictions when the SGCS values are lower than t_{SGCS}), and FN represents the number of false negatives (incorrect predictions when the SGCS values are lower than t_{SGCS}). The classification accuracy can be expressed as:

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}. \quad (18)$$

For precision and recall, the higher their values, the better the classification performance. As previously indicated, even when all predicted results are labeled as "True" in Figure 3 with the SGCS threshold t_{SGCS} set to 0.6, the classification accuracy ACC remains notably high at 88%. In this case, the precision P is 1, and the recall R is 0. This low recall R indicates that none of the actual low-quality feedback samples are being identified by the prediction method, i.e., $\text{TN} = 0$. Although the prediction method is highly precise in its positive classifications, it completely fails to identify any true instances of low-quality feedback, significantly undermining the transmission reliability.

- The evaluation metric ACC reflects the difference between the reconstructed CSI samples at the UE and BS using the proxy and original decoders, respectively. As mentioned before, the UE needs to generate CQI based on the original CSI $\mathbf{H}$ and the implicit CSI $\hat{\mathbf{V}}'$ generated at the UE. If the difference between the implicit CSI at the UE and BS, that is, $\hat{\mathbf{V}}'$ and $\hat{\mathbf{V}}$, is large, the CQI selected by the UE may not match the practical transmission condition. Therefore, the SGCS (represented as ρ_{proxy}^2) between the implicit CSI reconstructed at the UE and BS (i.e., $\hat{\mathbf{V}}'$ and $\hat{\mathbf{V}}$) should be adopted to assess the proxy model's performance and serves as the third evaluation metric. The definition of ρ_{proxy}^2 aligns with the formula presented in (6) and is omitted here to conserve space.

V. NUMERICAL SIMULATION RESULTS

In this section, we first outline the simulation settings, including channel generation and NN training. We then evaluate and discuss the proposed performance monitoring method for reliable AI-based CSI feedback in practical scenarios.

TABLE III

MASSIVE MIMO CHANNEL GENERATION PARAMETERS [47]

Parameter	Value
Carrier frequency	4 GHz
Bandwidth	10 MHz
Subcarrier spacing	15 KHz
RB number	52
Subband number	13
BS antenna number	32
UE antenna number	4
Channel model	3GPP UMi (80% indoor + 20% outdoor)
Sample slot interval	100
UE speed	indoor: 3 km/h; outdoor: 30 km/h
Rank	1
Channel estimation	ideal
UE number	3,000
Slot number	200

A. Simulation Settings

1) *Channel Generation*: An open channel dataset [47], complying with the 3GPP 38.901 channel model [48], is used to evaluate the proposed method. This dataset [47] includes 30 different sets of channel data, each corresponding to distinct channel models, and is utilized in the study of AI-based CSI feedback. Specifically, we employ the 25th dataset, labeled WLI_1_1025_UMi_52RB_3GPP_R18AI (referred to as Dataset A), for our analysis. The carrier frequency, bandwidth, and subcarrier spacing are set at 4 GHz, 10 MHz, and 15 KHz, respectively. The resource and subband numbers are 52 and 13, respectively. Both the BS and UE are equipped with 32 and 4 antennas, respectively. The 3GPP urban micro (UMi) channel model is applied, with UE speeds of 3 km/h for indoor and 30 km/h for outdoor scenarios. The channel rank is considered to be one. Further details about channel generation are available in Table III. A total of 600,000 CSI samples are generated, each containing $32 \times 13 \times 2 = 832$ real numbers in the implicit channel $\mathbf{V}$. The dataset is divided into training, validation, and test parts, comprising 80%, 10%, and 10% of the samples, respectively. The dataset (referred to as Dataset B) used in Section III-C for generalization evaluation differs from Dataset A. The primary settings remain consistent with those mentioned above, hence details are not reiterated. However, Dataset B employs a different channel model. This dataset, identified as WLI_1_1029_UMa_52RB_outdoor_nlos_3GPP_R18AI in [47], focuses on an urban macro (UMa) and non-line-of-sight (NLOS) scenario. A total of 60,000 CSI samples are randomly selected from the entire dataset for generalization evaluation.

2) *NN Training*: All simulations are conducted on an Nvidia DGX-2 workstation using TensorFlow (version: 2.13.0). The training optimizer utilized in this study is adaptive moment estimation [49]. The autoencoder is first trained using

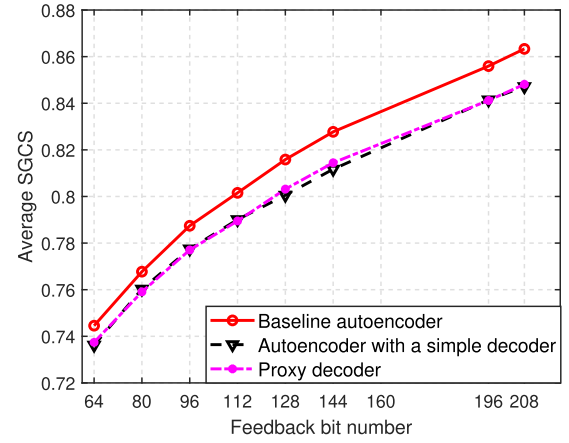


Fig. 12. Average SGCS ρ^2 versus feedback bit number N_{bits} . The performance of the proxy decoder is comparable to that of the autoencoder trained using an end-to-end approach.

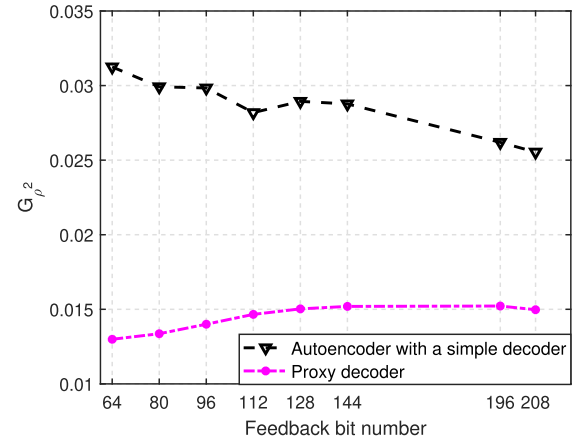


Fig. 13. Average gap G_{ρ^2} between the actual and predicted accuracy versus feedback bit number N_{bits} for the proxy decoder. The gap achieved by the proxy-based method is minimal.

an end-to-end approach, with a batch size of 256, a learning rate of 0.0001, and a training duration of 100 epochs. The quantization bit number, denoted as B , is consistently set to two in all simulations. The training hyperparameters for the proxy NN model are identical to those of the autoencoder during the end-to-end training phase.

B. Simulation Results

In this subsection, we first evaluate the monitoring accuracy of the proposed proxy model, followed by a complexity analysis.

1) *Average SGCS Performance of the Proposed Proxy Model*: In this section, we assess the average SGCS performance of the proxy NN model. Figure 12 illustrates the relationship between average SGCS ρ^2 and feedback bit number N_{bits} . The term “Autoencoder with a simple decoder” employs a decoder akin to the proxy NN, but it is jointly trained with the encoder in an end-to-end manner. As anticipated, the high-complexity autoencoder consistently outperforms its low-complexity counterpart, with the performance gap widening as the feedback bit number increases. Notably, when the feedback bit number is extremely limited, the feedback accuracy remains low for all methods. However, it is surprising to observe that the proxy decoder,

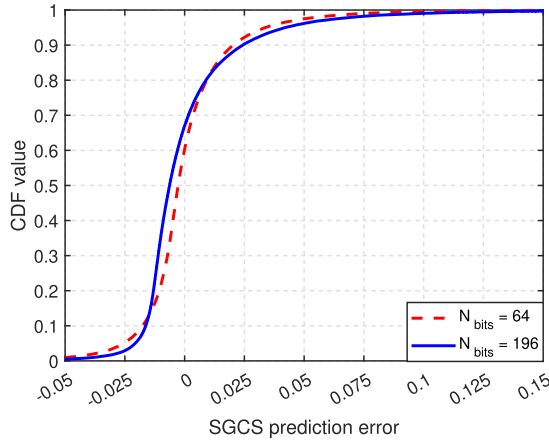


Fig. 14. CDF values of the SGCS prediction error for feedback bit numbers N_{bits} set at 64 and 196. Over 95% of samples exhibit prediction errors lower than 0.05.

despite not being jointly trained with the original encoder, demonstrates comparable performance to the autoencoder trained in an end-to-end fashion. This finding indicates the potential for designing and independently training new decoders, separate from pre-trained encoders. As a result, BS manufacturers could update the decoder independently of UE manufacturers.

2) Monitoring Accuracy of the Proposed Proxy Model:

In this section, we evaluate the proposed proxy-based performance monitoring method for reliable AI-based CSI feedback using different evaluation metrics introduced in Section IV-C.3. For comparison, we continue to use the “Autoencoder with a simple decoder” as a benchmark, as in Section V-B.1.

First, Figure 13 displays the average gap G_{ρ^2} between the actual and predicted accuracies as a function of the feedback bit number N_{bits} for the proxy decoder. The gap achieved by the proposed proxy-based method is minimal, typically around or below 0.015, a value that can be considered negligible. This gap remains relatively constant across different feedback bit numbers, indicating the reliability of the proxy-based method. Notably, this gap is about half of that achieved by the benchmark. In contrast, the benchmark’s gap diminishes with an increase in feedback bits. When a sufficient number of feedback bits are available, the CSI samples reconstructed by both the original powerful autoencoder and the autoencoder with a lightweight decoder are of high quality and closely resemble the ground truth, i.e., perfect CSI. As a result, the disparity between the reconstructed CSI samples is minimal. However, by leveraging the knowledge learned from the original powerful decoder, the proxy model proposed in this study can produce CSI comparable to that of the original decoder, irrespective of the number of feedback bits.

Then, Figure 14 shows the CDF values of the SGCS prediction error ($\rho_i^2 - \hat{\rho}_i^2 - \rho_d^2$) for feedback bit numbers of 64 and 196. Over 95% of the samples exhibit prediction errors lower than 0.05. Interestingly, more than 60% of the CSI samples have negative SGCS prediction errors, indicating the proxy model’s effectiveness, especially for challenging samples that require a more advanced decoder for reconstruction. The inclusion of ρ_d^2 in the SGCS prediction

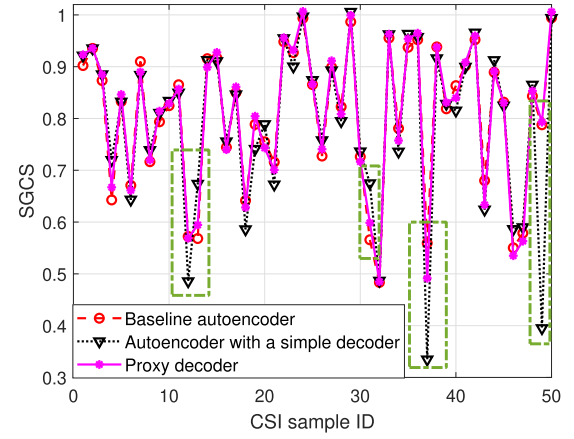


Fig. 15. SGCS values of 50 distinct CSI samples obtained using various methods, with a feedback bit number of $N_{\text{bits}} = 144$.

error calculation suggests that the SGCS values for most CSI samples might exceed those obtained by the original powerful decoder, i.e., $\hat{\rho}_i^2 + \rho_d^2 > \rho_i^2$.

Furthermore, Figure 15 displays the SGCS values of 50 distinct CSI samples obtained using various methods with a feedback bit number of 144. Four green dotted boxes highlight samples with significant SGCS prediction errors. The SGCS gaps between the original and proxy decoders are acceptable, particularly when compared to methods without knowledge distillation. The proxy decoder successfully emulates the mapping function of the original decoder, leading to high-quality SGCS predictions.

Next, we evaluate the classification performance of the proxy decoder using Figs. 16 and 17, which display the classification precision P and recall R , respectively, of the proxy decoder and the baseline against various SGCS thresholds t_{SGCS} with the feedback bit number N_{bits} set at 112. The precision values of both the proposed proxy decoder and the baseline method exceed 0.96 across all SGCS thresholds, with the proxy decoder slightly outperforming the baseline. As the SGCS threshold increases, the precision values decrease. A higher SGCS threshold implies that more CSI reconstructions fail to meet the accuracy requirement, making the classification task more challenging. The recall values, crucial for identifying low-quality feedback samples, are illustrated in Figure 17. Recall is a more critical metric than precision in the performance monitoring of AI-based CSI feedback, as the primary objective is to detect low-quality feedback samples to ensure reliable communication transmission. At a low SGCS threshold (e.g., $t_{\text{SGCS}} = 0.5$), most CSI samples meet the accuracy requirement, making it difficult to identify low-quality reconstructions. However, even under these conditions, the proposed method successfully identifies over 85% of low-quality reconstructions, significantly reducing the risk of poor-quality transmissions. When the SGCS threshold reaches 0.7, the method detects nearly 95% of inadequate CSI reconstructions. The proposed method consistently outperforms the baseline across all threshold values, particularly at lower thresholds.

Finally, we assess the disparity between the CSI reconstructions at the UE and BS using the proxy and original decoders.

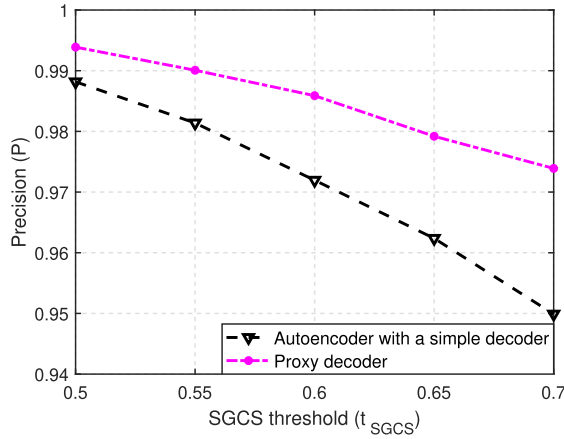


Fig. 16. Classification precision (P) of the proxy decoder and baseline against various SGCS thresholds.

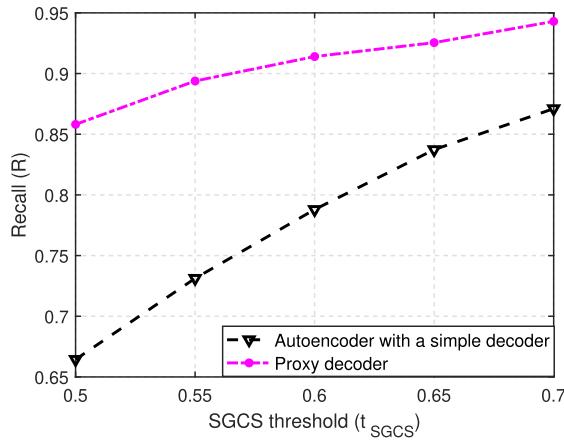


Fig. 17. Classification recall (R) of the proxy decoder and baseline against various SGCS thresholds t_{SGCS} .

Figure 18 compares the average SGCS ρ^2 between the reconstructed CSI obtained by the original and performance monitoring decoders against the feedback bit number N_{bits} . While the baseline method shows similar performance to the proxy decoder, as depicted in Figure 12, its reconstructed CSI output substantially differs from that of the original decoder. For example, at 112 feedback bits, the SGCS between the UE and BS reconstructed CSI using the baseline and original decoders is only 0.844, significantly lower than the 0.975 achieved by the proxy decoder. The consistently high SGCS values between the reconstructed CSI samples using the proxy and original decoders ensure high-performance CQI generation.

3) *Complexity Analysis:* As discussed in Section IV-C.1, the original decoder comprises six Transformer blocks to achieve high-quality CSI reconstruction. Due to considerations of complexity, the proxy decoder employs just a single Transformer block. Since the complexity of other operations in the decoder (such as dequantization, the FC layer, reshaping, etc.) is significantly lower than that of the Transformer block, the complexity analysis primarily focuses on the Transformer block. Consequently, the complexity of the proxy decoder is estimated to be approximately 14% of that of the original decoder.

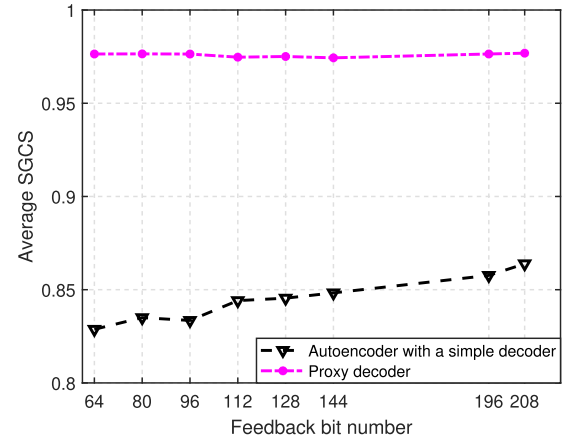


Fig. 18. Average SGCS ρ^2 between the reconstructed CSI obtained by the original decoder and the decoder used in performance monitoring, plotted against the feedback bit number N_{bits} .

VI. CONCLUSION

In this study, we introduced an efficient performance monitoring framework for reliable AI-based CSI feedback in practical scenarios. Our research stands out by focusing on monitoring the performance of AI-based CSI feedback, not just improving its accuracy through various AI methods and correlations. We began with comprehensive analyses of AI-based CSI feedback, adhering to 3GPP standard channel settings and highlighting the importance of performance monitoring in this context. Building on these insights, we explored a practical framework for AI-based CSI feedback. The core concept involves employing a lightweight proxy decoder to estimate the CSI reconstruction accuracy, effectively emulating the function of the original, more powerful decoder. We applied knowledge distillation to train the proxy decoder, enabling it to accurately replicate the mapping function of the original decoder. Additionally, we introduced several evaluation metrics for performance monitoring. The simulation results confirm that our proposed proxy-based performance monitoring strategy accurately predicts the feedback accuracy of AI-based CSI feedback with reduced complexity and accurately distinguishes low-quality feedback samples with a detection rate of nearly 95%, thereby preventing unreliable transmission. Moreover, the CSI reconstructed by the proxy decoder closely aligns with that produced by the original decoder, ensuring the generation of high-quality CQI.

Although the proposed proxy-based performance monitoring framework has shown promising results, some extensive challenges merit more in-depth exploration in the future. First, the design of a general proxy decoder like [39], which can accommodate varying feedback bits, is crucial. Second, meticulous consideration is needed in devising an approach regarding the feedback strategy change. Furthermore, it is worthwhile to investigate the choice of an appropriate SGCS threshold t_{SGCS} .

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